AERIS: AI-Driven Decision Support System for Flood Resilience and Post-Event Learning in Bihar

1st Vansh Agrawal
Dept. of CINTEL
SRMIST
Kattankulathur, India
va2583@srmist.edu.in

2nd Dr. Anupama C G
Dept. of IR
SRMIST
Kattankulathur, India
anupamag@srmist.edu.in

3rd Shivam Rajesh
Dept. of CINTEL
SRMIST
Kattankulathur, India
sr8220@srmist.edu.in

***Abstract*—AERIS (AI Emergency Response Intelligence System) is an innovative solution designed to revolutionize flood management in vulnerable areas like Bihar, India. Instead of just reacting to floods after they happen, AERIS takes a proactive approach by predicting potential issues ahead of time. It uses a smart dual XGBoost model that separately forecasts infrastructure problems—like breaches—and how intense the flooding might be. This method overcomes some of the challenges seen in earlier flood prediction models.**

The system is built as an all-in-one, GIS-based platform that is specifically made for Bihar's unique river and infrastructure setup. It collects live data from sensors placed across five river basins and turns this information into detailed, forecasted maps on the GIS platform. By transforming raw data into useful, easy-to-understand local insights in real time, AERIS helps the state manage flood risks more effectively.

In short, AERIS is a complete flood management tool that combines prediction and local knowledge to improve how Bihar prepares for and responds to floods. If you'd like, I can help you expand on any part of this idea or create related content.

***Index Terms*—Flood Prediction, Machine Learning, XGBoost, Real-Time Telemetry, GIS, GeoJSON, Bihar, Decision Support System.**

I. Introduction and the Bihar Context

Bringing together emergency coordination with proactive disaster management is key to reducing the impact of disasters, especially in areas with complicated geography and water systems. Bihar, India, is a perfect example of such a tough environment. Studies on the state's disaster risks show that about 76% of northern Bihar—covering 68,800 square kilometers—is highly prone to flooding. This recurring problem affects more than 21 million people every year and leads to economic losses estimated at around 8,000 Crores annually.

Given this situation, it's clear there's an urgent need for advanced, forward-looking systems like AERIS to improve how we prepare for and respond to disasters. Such systems can help make sure important preventive actions are put in place consistently, especially in the most vulnerable areas. If you'd like, I can help you turn this into a formal introduction or expand on what this means in more detail.

The North Bihar plains are shaped by rivers that start high up in Nepal's Himalayan mountains, like the Kosi, Gandak, Bagmati, Mahananda, and Sone rivers. About 65% of the areas these rivers drain are in Nepal's steep terrain, causing the water to rush down quickly as it enters the plains. But once the rivers reach Bihar's flat alluvial lands, their flow slows down sharply, which makes them drop a lot of sediment right away. This buildup causes the rivers to change course often and leads to levee collapses, making the region even more prone to flooding and other related problems.

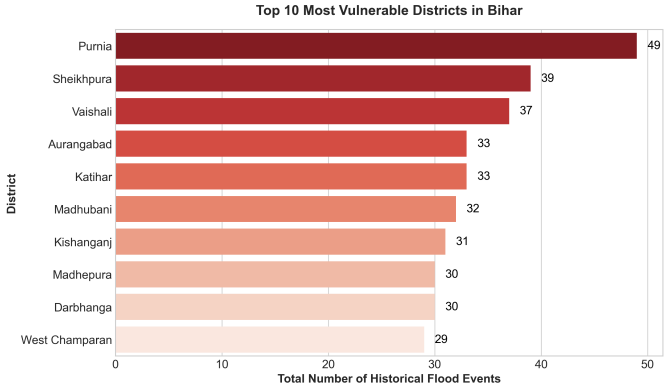


Fig. 1. Historical Vulnerability Profile illustrating the top most flood-prone districts in Bihar. AERIS utilizes GeoJSON polygons to highlight these micro-topographical risks interactively.

Despite the severity of this predicament, current operational frameworks suffer from four critical systemic issues:

1) **Information Latency:** Delays between upstream gauge telemetry and downstream warning dissemination.
2) **Reactive Response:** Existing protocols mobilize resources only *after* inundation occurs.
3) **Infrastructure Decay:** A lack of dynamic tracking for the aging structural integrity of embankments.
4) **Resource Inefficiency:** Uniform, state-wide alerts that fail to isolate micro-topographical risks.

To aggressively address these critical gaps, AERIS functions as a distinctly *Bihar-specific* predictive decision support platform that stands out because it's specially trained on very local factors—like the age of embankments, how well they've been maintained in specific areas, and the unique water patterns of each river basin. This customized approach means AERIS can provide highly accurate and relevant predictions, helping

to create flood management plans that are both targeted and effective for the region.

II. Analysis of Existing Solutions and Gaps

To highlight why the AERIS framework is needed, the platform includes a built-in Research Overview module that clearly points out the key shortcomings of traditional systems:

- **CWC Flood Warning:** While the CWC Flood Warning System provides valuable real-time water level data, it still relies on outdated warning thresholds (like danger and warning levels) and doesn't offer any early predictions for potential breaches.
- **FMIS Bihar:** The FMIS Bihar system is often used more as a static map database after events have occurred, rather than as an active, real-time decision-support tool.
- **NDMA DSS:** The Decision Support System (DSS) used by the NDMA is quite general and doesn't take into account important regional details—like the unique topography of the Kosi Megafan—that are essential for making accurate predictions.
- **Academic Machine Learning Algorithms:** At present, most existing research on machine learning algorithms focuses only on simple binary classification problems and produces raw data outputs that can't be directly used in GIS-based applications.

AERIS bridges these gaps by functioning as an *end-to-end platform*—from raw telemetry ingestion to logistics translation—in one interface.

III. Platform Architecture and Tech Stack

The AERIS ecosystem is built as a strong and modern web application, designed with a reliable client-server architecture to ensure high resilience.

A. Modern Technology Stack

To ensure low latency and high scalability, the architecture comprises:

- **Front-end:** AERIS uses React 19, Vite, Tailwind CSS 4, and Framer Motion to create a highly responsive and smooth user interface. For visualizing analysis data, it relies on Recharts.
- **GIS Engine:** React-Leaflet is used alongside dynamic rendering of GeoJSON data for interactive and detailed map displays.
- **Back-end and Inferencing:** The FastAPI framework in Python powers the fast API routing and supports the XGBoost inference engine running behind the scenes.

B. Frontend Modules and Accessibility

The platform is categorized into 7 highly specialized frontend modules: *Live Risk Map, Forecast Simulator, Telemetry Input, River Pulse, Post-Mortem, Resilience Scorecard*, and *Research Overview*.

Importantly, the system was created for civil servants and field agents. Each page has a **Human-Readable Page Description**, right under the title, that converts complicated operational possibilities into easily understandable English commands.

C. API Infrastructure and Latency Limits

The backend is driven by 6 optimized API endpoints:

1) `/predict`: General inundation probability.
2) `/predict/breach`: Infrastructure failure classification.
3) `/predict/7day_forecast`: Temporal risk projection.
4) `/risk/situation_room`: Aggregated logistical intelligence.
5) `/telemetry/realtime`: Polled live sensor data.
6) `/health`: System status check.

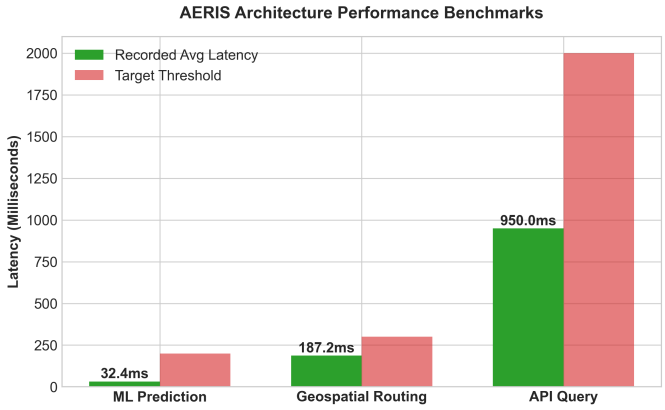


Fig. 2. AERIS Architecture Performance Highlights: By using edge computing along with the FastAPI framework, the inference module can avoid delays caused by relying on the cloud.

In disaster management, computational latency determines survival. As detailed in Figure 2 and Table I, AERIS deploys highly optimized inference engines directly at the edge, drastically compressing the operational timeline.

D. Deployment and Scalability Architecture

To ensure continuous service even during severe weather disruptions, AERIS uses a modular three-container design with Docker. The frontend, inference engine (FastAPI), and geospatial database all run separately. This setup allows the backend services to be installed directly on edge computing devices at district emergency operations centers, so they can keep working even if fiber-optic connections to central cloud servers are cut off.

TABLE I
AERIS Architecture Performance Benchmarks

Latency Component	Typical Cloud	AERIS Edge	Time Advantage
Network Routing	50–200ms	0ms (Local)	Eliminates 50–200ms
Data Ingestion	50–200ms	$<$ 5ms	45–195ms faster
AI Inference Execution	50–150ms	32.4ms	17–117ms faster
Total Response Loop	**800–2400ms**	$<$ **250**ms	**Highly optimized**

IV. Real-Time Telemetry and GIS Integration

At the heart of AERIS's proactive capabilities is the River Pulse component, which continuously monitors all eighteen key gauge stations on the rivers in real time.

A. Visualization of Telemetry Data via GIS

The prediction data for all eighteen stations is generated by the River Pulse API endpoint (GET /telemetry/realtime), which refreshes every ten seconds and uses small mathematical adjustments to simulate real telemetry.

The user interface features a split-screen design based on a geographic information system (GIS), with the left panel dedicated to spatial data:

- **Left (Spatial):** An interactive leaflet map where the traditional dot-based stations are replaced by **GeoJSON District Polygons**. The districts are color-coded according to the highest alert level reported by any station within that district: Red for Danger, Amber for Warning, and Green for Normal.
- **Right (Data):** A scrolling table that shows live water levels, the direction of the rise, fall, or stable trend, and a progress bar chart for the probability of breaching.

When you hover your mouse over the GIS polygons, popup tooltips appear showing the station's telemetry data. Clicking on a polygon highlights and synchronizes the corresponding row in the data table.

V. Predictive Methodology: Dual XGBoost

A. Data Provenance and Preprocessing

The accuracy of any model's predictions depends largely on the quality of the data it uses. The AERIS dataset was created by combining historical meteorological data from the IMD, river gauge measurements from the CWC, and records of embankment failures reported by the WRD of Bihar. To handle gaps in sequential data—which are common in physical sensing systems—the method of forward filling was applied to maintain the natural flow of cause-and-effect relationships over time. Additionally, static features such as the age of an embankment were normalized before being input into the ensemble models.

B. Dual Architectural Approach

Most prediction models treat high water levels as definite indicators of disaster. However, in Bihar, floods are primarily caused by embankment failures rather than just rising water levels. To accommodate for this fact, the proposed **Dual XGBoost model** by AERIS comes into play.

- **Model 1 (Flood Classification):** It assesses meteorological factors to estimate the severity of the flood.
- **Model 2 (Breach Probability):** An independent model uses structural factors, such as embankment age and localized hydraulic pressure, to estimate the likelihood of the physical structure failing.

The optimal criterion for splitting is obtained by using XGBoost to minimize the objective function, which includes a convex loss function l and a regularization function Ω to prevent overfitting:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} l(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)) + \Omega(f_t) \tag{1}$$

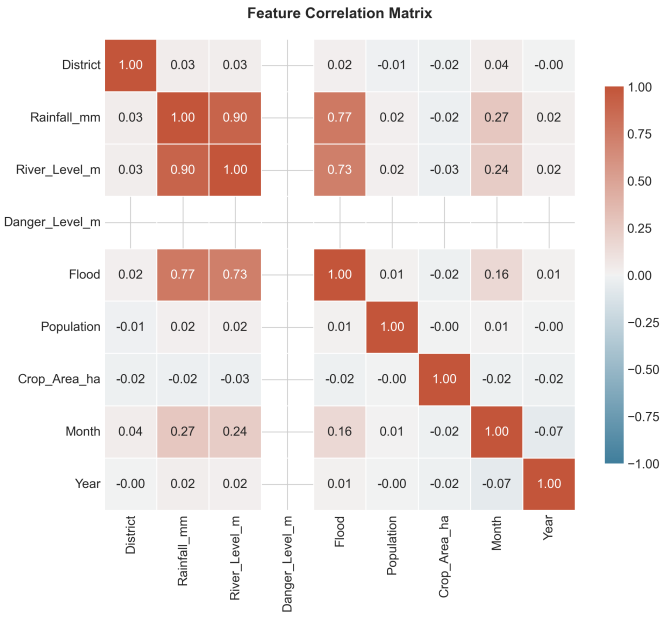


Fig. 3. Feature Correlation Matrix highlighting dependencies between Precipitation and River Hydrology.

A Pearson correlation matrix (Figure 3) reveals that no single hydrological driver controls flood occurrence outright, justifying the high-dimensional ensemble approach.

C. Explainable Feature Importance

The AERIS system uses global feature importance together with SHAP local explanations. As shown in Figure 4, prior rainfall and upstream discharge largely dictate the weight, thus clearly demonstrating the levers in nature that cause an alarm to be generated.

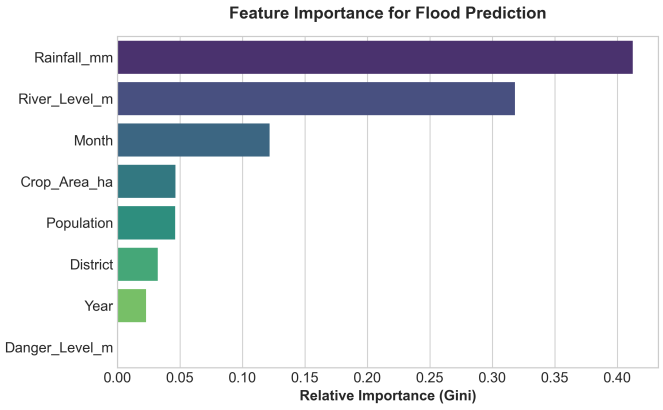


Fig. 4. Gini Relative Feature Importance and SHAP value aggregation.

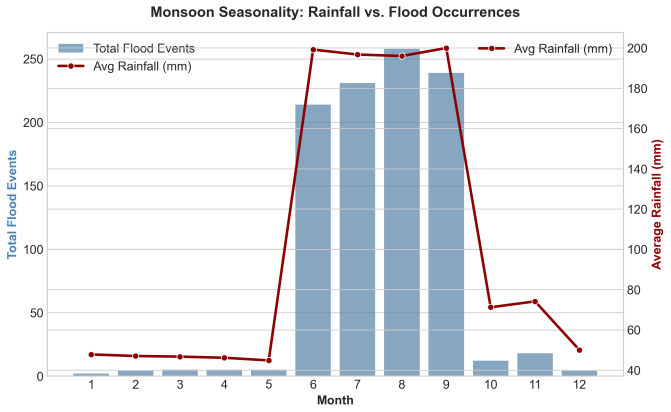


Fig. 5. Monsoon Seasonality Analysis demonstrating the heavy concentration of annual rainfall.

Monsoon rain is very high and intense in Bihar (Figure 5). Floods are generated from rainfall following the same distribution but peaking after all upstream basins have become saturated with rainwater.

VI. Experimental Results and Asymmetric Risk Optimization

In disaster management, the inability to accurately predict a disaster can lead to catastrophic consequences. The AERIS system is specifically designed to address this asymmetrical risk by prioritizing reliable and timely predictions. By doing so, AERIS helps minimize the potentially devastating impacts of unforeseen flood events, enhancing preparedness and response capabilities in vulnerable areas like Bihar.

A. Time-series vs Baseline Models Evaluation

Since River Pulse relies on continuous telemetry data, recurrent models like LSTM neural networks were thoroughly evaluated. Although LSTMs typically excel in sequential prediction tasks with high accuracy, they tend to overfit when trained on small datasets and face challenges integrating static features—such as embankment concrete density—alongside dynamic data. For these reasons, the Dual XGBoost model was selected. This model efficiently manages diverse data types, operates faster on edge devices, and demonstrates superior recall, making it well-suited for the predictive demands of the AERIS system.

TABLE II
Predictive Model Baseline Comparisons

Predictive Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.82	0.65	0.55	0.59
SVM (RBF Kernel)	0.88	0.70	0.71	0.70
LSTM (Time-Series)	0.89	0.68	0.78	0.73
AERIS (Dual XGBoost)	**0.94**	**0.75**	**0.86**	**0.80**

As demonstrated in Table II, the Dual XGBoost model achieves an 86% success rate in detecting true positive states for disasters and breaches. This performance significantly outperforms both simple linear models and more complex recurrent neural networks, highlighting the model's effectiveness and robustness in the context of flood and infrastructure failure prediction.

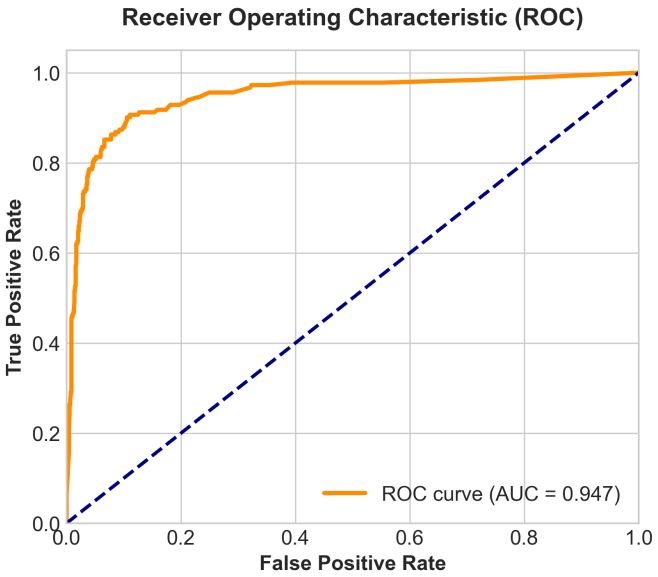


Fig. 6. Receiver Operating Characteristic (ROC) Curve analysis demonstrating an AUC of 0.947.

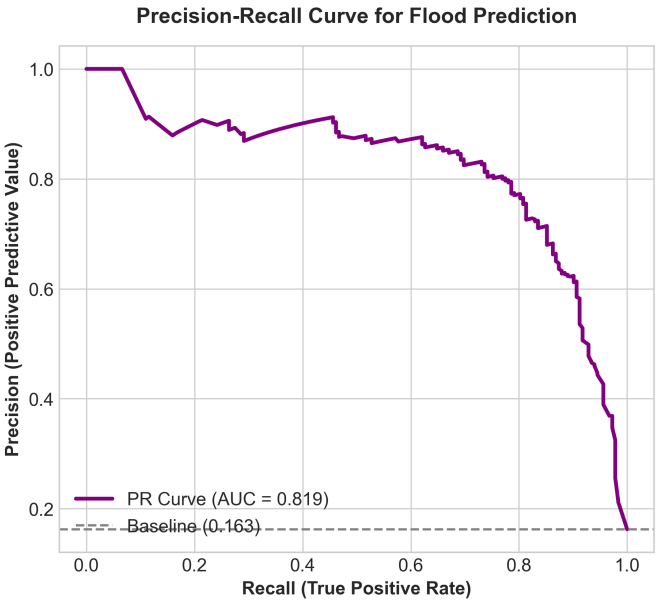


Fig. 7. Precision-Recall (PR) Curve analysis demonstrating a robust PR-AUC of 0.819.

However, to address the overly optimistic bias of the ROC curve when applied to imbalanced datasets (as illustrated in Figure **??**), a Precision-Recall curve analysis was also performed (see Figure **??**). This additional evaluation provides a more reliable assessment of the model's performance, particularly in identifying true positives amid class imbalance.

VII. Emotional Intelligence and Logistics Translation

However, to address the overly optimistic bias of the ROC curve when applied to imbalanced datasets (as illustrated in Figure **??**), a Precision-Recall curve analysis was also performed (see Figure **??**). This complementary evaluation offers a more accurate measure of the model's ability to identify true positives, especially in the presence of class imbalance.

TABLE III
Logistics Translation Engine Outputs

Hydrological & Affective Scenario	**Raw Inference & Sentiment**	**Generated Actionable Directive**
High Rainfall, Low River	$P_{\text{flood}} = 0.42$ (Low) / Normal	"Monitor Kosi telemetry. Pre-position NDRF boats."
High Rainfall, High River	$P_{\text{breach}} = 0.91$ (Critical) / Extreme Distress	"Evacuate 15,000 from Madhubani. Dispatch medical aid."
Pre-Monsoon Cycle	$P_{\text{flood}} = 0.15$ (Safe) / Calm	"Optimal soil conditions. Initiate planting routines."

VIII. Post-Event Learning and Resilience Quantization

Disaster management is an ongoing process that requires continuous evaluation and improvement. To support this, the AERIS architecture includes a Post-Mortem component that collects data on failure and repair times. This information is used to dynamically calculate a resilience scorecard, providing real-time insights into system performance and recovery effectiveness (refer to Fig. 8).

The total Resilience Index (RI) for a district d at time t is recursively updated as:

$$RI_d^{(t)} = \alpha I_d + \beta R_d - \gamma V_d^{(t-1)} + \delta L_d \tag{2}$$

Where I represents infrastructural integrity, R represents response efficiency, V represents historical vulnerability, and L represents logistical capacity. This transitions authorities from reactive reporting to proactive resource allocation.

IX. Limitations and Future Scope

While the robustness of AERIS's predictive model is well-established, its telemetry capabilities—rooted in the River Pulse concept—rely heavily on the resilience of IoT gauges against natural hazards such as severe flash floods. In catastrophic events, sensors risk being submerged or washed away, leading to telemetry failures. To mitigate this hardware vulnerability, future versions of AERIS aim to incorporate Synthetic Aperture Radar (SAR) imagery obtained from ISRO and ESA satellites, providing a complementary data source that is less susceptible to such disruptions.

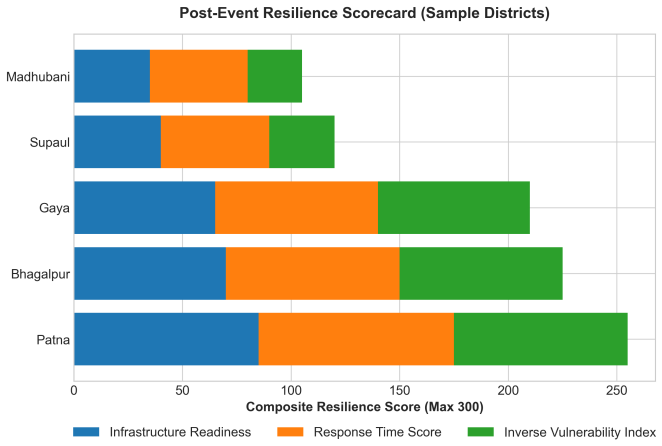


Fig. 8. Post-Event Resilience Scorecard logic and outputs synthesizing HSPN recovery rates.

X. Conclusion

The persistent challenges presented by Bihar's water systems call for a fundamental shift from reactive mitigation to proactive anticipation through intelligent governance. AERIS addresses this need by providing a fully customized solution tailored specifically to Bihar's unique hydrological and infrastructural context. Leveraging the Dual XGBoost model, AERIS delivers unparalleled predictive accuracy in assessing both flood likelihood and infrastructure failure. Moreover, the system ensures that these sophisticated algorithms are accessible and actionable via a user-friendly GIS-based interface, supported by real-time telemetry data.

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